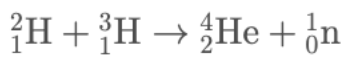


Section: Particles and Waves

Topic: Nuclear Reactions

The following nuclear reaction occurs in the core of a star:



(a) Confirm that the reaction is balanced in terms of mass number and atomic number.

Check **mass numbers** (top):

$$2 + 3 = 5 \quad \text{and} \quad 4 + 1 = 5 \quad \Rightarrow \quad \text{balanced}$$

Check **atomic numbers** (bottom):

$$1 + 1 = 2 \quad \text{and} \quad 2 + 0 = 2 \quad \Rightarrow \quad \text{balanced}$$

✔ The reaction is balanced in both mass number and atomic number.

(b) This is a fusion reaction.

State one difference between **nuclear fusion** and **nuclear fission**.

✔ **Fusion** involves:

Two **light nuclei** combining to form a **heavier nucleus**

✔ **Fission** involves:

A **heavy nucleus** splitting into two or more **lighter nuclei**

(c) Explain why a very high temperature is required for fusion reactions to occur.

✔ The nuclei are **positively charged**, so they **repel** each other due to electrostatic (Coulomb) repulsion.

A **very high temperature** gives them **enough kinetic energy** to overcome this repulsive force and get close enough for the **strong nuclear force** to bind them together.

🧠 Revision Tips

- Always check nuclear reactions for conservation of:
 - **Mass number** (top number)
 - **Atomic number** (bottom number)
- Fusion = **joining** of light nuclei
- Fission = **splitting** of heavy nuclei
- Fusion requires extreme conditions (like in the Sun) due to **electrostatic repulsion**